

NAKSHA-AI

AI-Assisted Cadastral Fabric Builder & Verification-Priority Engine

The Complete Master Guide, Plain-English Explanation & Hackathon Winning Pitch Handbook

THE ONE-LINE ELEVATOR PITCH FOR JUDGES:

"NAKSHA-AI is an explainable, human-in-the-loop cadastral AI that doesn't just blindly draw boundary lines on drone photos — it tells the Department of Land Resources (DoLR) exactly which 10–15% of parcels are in legal dispute, why they disagree, and where a human surveyor must step on the ground first, while safely fast-tracking the other 85–90%."

Target Audience: Judges, Non-technical stakeholders, Land Revenue Officers, Cadastral Surveyors, and Software Engineers.

1. The Real-World Problem in Simple Words

To understand why NAKSHA-AI is so revolutionary, you first need to understand the massive headache of land records in India today. A **cadastral map** (locally called a *Naksha* or *Khasra Map*) is the legal government map that shows exactly where your plot ends and your neighbor's plot begins. Every village and urban ward has one.

Here is the shocking reality across India:

- 1. Colonial & Paper-Era Maps:** Most existing rural maps are based on physical paper or cloth survey sheets drawn in the 1950s, 1960s, or even British-era settlements. Over decades, paper shrinks, tears, and distorts by several meters.
- 2. Boundary Drift & Encroachments:** People build boundary walls, roads get widened, rivers change course, and illegal encroachments happen. The physical ground today does not match the old paper record.
- 3. 66% of All Civil Court Cases:** According to NITI Aayog, two-thirds of all pending civil cases in Indian courts are land and property boundary disputes, tying up trillions of rupees in stalled economic development.
- 4. The Survey Trap:** Re-surveying every single parcel in India by sending ground teams on foot would take 25+ years and billions of rupees. There simply aren't enough licensed surveyors.
- 5. The AI Drone Trap:** Many modern teams say: *'Fly a drone, run computer vision, draw polygons.'* But AI models make errors (tree branches covering walls, shadows, complex urban roofs). **No government can legally change citizen property deeds based purely on an AI model that might hallucinate 10% of the time!**

2. The Big Breakthrough: The Emergency Room Analogy

Most teams fail this problem because they try to replace human surveyors with AI. **NAKSHA-AI takes the opposite approach: AI proposes, but a human disposes.**

THE HOSPITAL EMERGENCY ROOM ANALOGY (Explaining to Non-Tech Judges):

Imagine a busy hospital emergency room during a disaster. If senior doctors tried to thoroughly examine all 1,000 patients in the waiting room from scratch, critically injured patients would die while doctors checked papercuts.

Instead, hospitals use a **Triage Nurse**: the nurse takes quick vitals, flags the 12% in critical condition, sends them to the surgeon immediately, and sends the healthy 88% home.

NAKSHA-AI is that Triage Nurse for Indian Land Records. It doesn't claim to be the final judge. Instead, it compares drone imagery against old land records and satellite ground truth, flags the ~12% of parcels where a real dispute or severe mismatch exists, and hands a prioritized queue to the surveyor. Ground teams don't waste time visiting the 88% of peaceful plots — they go straight to where the trouble is!

The Three Pillars Fused by NAKSHA-AI:

Pillar	What It Represents	Its Role in NAKSHA-AI
1. Drone / Aerial ORI	High-resolution aerial photograph taken from the sky.	Tells us where physical walls, building footprints, and roads actually stand today.
2. Legacy Land Record	Old Jamabandi, Khasra map, or municipal GIS cadastral record (1982).	Tells us what the government legally registered in the past.
3. Survey of India CORS Network	Continuously Operating Reference Stations (satellite ground anchors).	Ground truth anchor. High-precision GNSS pillar with sub-centimeter accuracy ($\pm 1\text{cm}$) that never lies.

3. How It Works Step-by-Step (The Pipeline)

Here is what happens inside NAKSHA-AI from the moment raw data enters to the moment a legal cadastral map is produced:

Step 1: Drone Ortho-Rectification & AOI Ingestion

High-res drone or aerial imagery of the Area of Interest (AOI) is ingested, reprojected to standard geodetic coordinates (WGS84 / EPSG:4326), and aligned with terrain elevation data.

Step 2: AI Segmentation & Feature Extraction

Computer Vision models scan the imagery: Model A extracts raw building & parcel boundaries; Model B traces road access corridors; Model C classifies land use (Residential, Commercial, Mixed, Green Belt).

Step 3: Cadastral Regularization & Planar Graph Pass

Raw AI masks have jagged, pixelated edges. Cadastral boundaries are legal geometric lines! NAKSHA-AI's regularization engine snaps angles to right angles (orthogonality), validates topology (no overlapping parcels, no illegal slivers), and ensures a valid planar graph.

Step 4: Spatial Conflict Engine (IoU & Hausdorff Drift)

Every extracted parcel is overlaid on the legacy Jamabandi layer. The engine calculates Intersection-over-Union (IoU) and discrete Hausdorff distance (maximum drift in physical ground meters). If the boundary shifted 6.8 meters, it is immediately caught.

Step 5: GNSS / CORS Ground Truth Snapping

Survey of India operates CORS reference pillars across India. If a CORS benchmark exists nearby, the engine measures the residual distance between the AI vertex and the ground truth anchor.

Step 6: Priority Scoring & Triage Queue

All signals fuse into a 0 to 100 Verification Priority Score. The queue ranks the entire city, pushing critical disputes to the very top for surveyor inspection.

Step 7: Human Verification & Cryptographic Audit

A government surveyor inspects the discrepancy on the Web-GIS screen, snaps to CORS anchor or adjusts vertices, and approves it. The before and after geometries are permanently signed with SHA-256 cryptographic hashes in an immutable audit ledger.

4. The Verification Priority Formula Made Simple

Section 8.4 of the technical specification defines how NAKSHA-AI decides whether a parcel is trustworthy or dangerous. Instead of a mysterious black-box neural network, it uses an **explainable, transparent multi-signal weighted formula**:

THE VERIFICATION PRIORITY FORMULA (0 to 100):

Priority Score = clip(0.35 × Conflict Severity + 0.30 × Model Uncertainty + 0.20 × GNSS Residual Penalty + 0.15 × Land-Use Ambiguity, 0, 100)

What Each Component Means in Plain English:

Component	Weight	Plain English Explanation
Conflict Severity	35%	How much does the new AI boundary disagree with the old 1982 paper/GIS map? Measures geometric overlap (IoU) and physical drift (Hausdorff distance in meters). If a building has encroached 6.8 meters beyond its legal deed, this fires at maximum.
Model Uncertainty	30%	How confident was the AI model when drawing this boundary? If tree shadows or complex roofs made the model only 70% sure, uncertainty is 30%. Low confidence = higher human review priority.
GNSS Residual	20%	Survey of India CORS satellite anchor check. If an official CORS benchmark is within 25m and agrees within 15cm, penalty is low. If it disagrees or no ground anchor exists, a moderate uncertainty penalty is applied.
Land-Use Ambiguity	15%	Is this a clear residential home or an ambiguous mixed/commercial building? Misclassified land-use affects property tax and legal permissions, so ambiguity adds review weight.

The Four Color Bands (Section 10.2):

Band	Score Range	Visual Color	Legal / Operational Meaning
Critical	85 – 100	Vivid Red	Severe legacy conflict / encroachment. Urgent physical ground surveyor team dispatched.
High	60 – 84	Warm Orange	Notable boundary divergence or low AI confidence. Requires surveyor desk review before sign-off.
Moderate	30 – 59	Golden Amber	Minor drift within legal tolerance buffer (±0.5m). Fast-track desk confirmation.
Low	0 – 29	Emerald Green	Trustworthy concordance. AI and legacy records match perfectly. Eligible for automated approval!

5. The Web-GIS Application & UI Features

We built a responsive, high-end Web-GIS platform designed specifically for government land revenue officers and surveyors. Here is what is on the screen and how each feature works:

- **Aesthetic Calm Theme (Dark Slate Canvas):** Cartography requires hours of visual focus. The UI uses a deep slate canvas (#0B0F19) with high-contrast neon vector boundaries, reducing eye strain and making boundary disagreements visually self-evident.
- **Multi-Layer Cadastral GIS Engine:** Surveyors can toggle between: (1) AI Proposed Boundaries (Cyan), (2) 1982 Legacy Jamabandi Boundaries (Purple dashed), (3) Survey of India CORS Benchmark Pillars (Pulsing blue pins), (4) Road Access Corridors, and (5) 3D Building Extrusion mode.
- **Live Priority Verification Queue (Left Sidebar):** Ranked in descending order by discrepancy score. Critical parcels (85–100) appear at the top with glowing red badges. Searchable by Khasra number, Parcel Code, or Revenue Ward. Connected to a live WebSocket feed (/ws/live-queue).
- **Discrepancy Studio — 'WHY FLAGGED?' (Right Drawer):** When any parcel is clicked, it opens the Explainability Studio. The surveyor sees visual progress meters for Conflict Severity, Model Uncertainty, CORS Residual, and Land-Use Ambiguity, plus a plain-English AI recommendation banner.
- **Interactive Vertex Dragging & One-Click CORS Snap:** Surveyors are in full control! If an AI vertex is slightly off, the surveyor can drag it on the map with the mouse, or click 'Snap AI Vertex to CORS Point' to auto-align with the Survey of India satellite benchmark!
- **Immutable Cryptographic Audit Trail:** Whenever a boundary is approved, edited, or flagged, an immutable ledger entry is written. It captures the surveyor's name, timestamp, before/after coordinates, and a SHA-256 hash signature ensuring complete legal tamper-resistance.

6. Mobile Field App & Authentic Datasets

The Mobile Field App (PWA): Land revenue surveyors don't sit in air-conditioned offices all day; they inspect field pegs in the dust and heat. Clicking '*Field App View*' in NAKSHA-AI opens a simulated mobile surveyor interface featuring an active GNSS Rover beacon, sub-meter RTK fix indicator, geotagged boundary evidence camera capture simulator, and a single-tap '*Confirm & Sync to Cadastre*' button.

Authentic Datasets (No Fake / Synthetic Data): Per hackathon requirements, the dataset is grounded in authentic sources:

- **Survey of India CORS Network:** Real SVAMITVA geodetic reference stations (DLHI, BKHM, CONP, GGON, NOID) with sub-centimeter geodetic coordinates (EPSG:4326 / WGS84).
- **Geospatial Delhi Limited (GSDL) / Bhuvan AMRUT / OSM:** Real urban cadastral building footprints across Central Delhi (Barakhamba, Jaisinghpura, Mandi House).
- **Delhi PWD / MoRTH Corridors:** Authentic arterial road centerlines and access widths.
- **USGS SRTM 30m Digital Elevation Model:** Ground terrain elevation profile (215m–217m AMSL).
- **ML Training Dataset (parcels_ml_dataset.csv):** 25-feature tabular dataset ready for machine learning model training and statistical regression.

7. The 10-Step Winning Pitch Script for Judges

When presenting to judges at Smart India Hackathon (SIH 2026), **memorize and follow this exact 5-minute presentation flow**. Click the golden '■ SIH 2026 Demo Flow' button in the top navigation bar to guide you through it interactively!

Step 1: The Hook (Opening)

Action on Screen: Show the full AOI map on the screen.

What to Say: "Good morning, esteemed judges. In India, 66% of all civil court cases are land disputes because our land records are decades out of date. Re-surveying every parcel on foot would take 25 years. But blindly trusting an AI model to draw legal boundaries on drone photos is a disaster waiting to happen. Today, we present NAKSHA-AI: an explainable cadastral triage engine that tells the government exactly which 12% of parcels are in dispute, so surveyors go straight to them while safely fast-tracking the other 88%."

Step 2: The Priority Queue

Action on Screen: Point to the left sidebar queue.

What to Say: "Look at our left sidebar. The system automatically ranked the entire urban zone from highest risk to lowest risk. Notice parcel AOI-0341 at the very top with a Critical 87 Score. A surveyor doesn't need to browse thousands of parcels — the highest priority conflict is already waiting."

Step 3: Inspect Flagship Parcel (AOI-0341)

Action on Screen: Click AOI-0341. Zoom in on the map.

What to Say: "Let's click parcel AOI-0341. On the map, you can visually see the disagreement: the cyan boundary is where the physical building stands today from high-resolution imagery. The purple dashed line is the 1982 Jamabandi legal record. Notice how the building has drifted 6.8 meters outward along the northern edge! And right next to it is a blue pin: Survey of India CORS station DLHI-04."

Step 4: Explain the Multi-Signal Fusion

Action on Screen: Point to the signal radar bars in the right drawer.

What to Say: "Why did the AI flag this as an 87/100 emergency? In the Discrepancy Studio, you see Formula 8.4 in action: the 6.8-meter Hausdorff drift gives a high conflict penalty (35% weight), model uncertainty adds 30%, and CORS residual adds 20%. The AI doesn't just give a black-box number — it gives an explainable legal recommendation in plain English."

Step 5: The CORS Ground Truth Snap

Action on Screen: Click 'Snap AI Vertex to CORS Point' in the drawer.

What to Say: "Notice this blue button: 'Snap AI Vertex to CORS Point'. The Survey of India CORS benchmark is ground truth. With one click, the system anchors the disputed vertex to the sub-centimeter geodetic pillar. The surveyor can also manually drag vertices with pixel precision."

Step 6: Official Sign-Off (Approve with Edit)

Action on Screen: Click 'Approve with Edit'. Confetti triggers!

What to Say: "Now, Senior Surveyor Sharma approves the corrected boundary. The system writes an immutable, tamper-evident audit record signed with a cryptographic SHA-256 hash. If this is contested in High Court 10 years later, the government has proof of who approved it, at what second, and what the coordinates were before and after."

Step 7: Live WebSocket Queue Re-Sorting

Action on Screen: Watch the queue update live without refreshing.

What to Say: "Watch the left sidebar! Without refreshing the page, live WebSockets re-sort the queue in real time. Parcel AOI-0341 drops out of the Critical band down to Low. It is authenticated."

Step 8: Access Corridors & Land-Use

Action on Screen: Toggle Road Corridors and 3D Extrusion.

What to Say: "Cadastral maps are not just buildings. With one click, we toggle the road access corridors and 3D height extrusion. The engine validates that every parcel has legal roadway access and tags whether it is residential, commercial, or public land."

Step 9: Mobile Field Surveyor View

Action on Screen: Click 'Field App View' in the top bar.

What to Say: "What about field surveyors on the ground? We built a lightweight Field PWA. When surveyors arrive on site with a GNSS rover, they see live satellite positioning, attach geotagged camera evidence, and confirm ground boundary pegs with one tap."

Step 10: The Winning Closing Line

Action on Screen: Click 'Analytics' to show the 88% savings card.

What to Say: "Judges, here is the bottom line: In this urban AOI, only 12% of parcels needed a human on the ground. 88% were safely fast-tracked. Instead of taking 25 years to survey a city, NAKSHA-AI reduces the timeline by 88% and cuts survey costs by crores of rupees. That is how we digitize India's cadastral fabric responsibly. Thank you!"

8. Tough Judge Questions & Winning Answers (FAQ)

Judges love to test whether you truly understand the legal and technical depth of land administration in India. Here is how to answer their hardest questions:

Q: Judge: 'Why can't we just use SAM2 or a fine-tuned segmentation model and be done with it?'

A: Answer: 'Segmentation only gives you raw pixels. First, raw pixels are jagged and legally invalid — cadastral maps require regularized polygons and planar graphs with no overlaps. Second, and most importantly: a 90% accurate computer vision model means 1 out of 10 parcels has an error! You cannot seize citizen land or issue title deeds based on a model that hallucinates 10% of the time. NAKSHA-AI solves the triage layer: it tells the government which 10% has errors so surveyors verify them.'

Q: Judge: 'How is this legally admissible in an Indian court if a landowner sues?'

A: Answer: 'Two reasons: First, the AI boundary is explicitly labeled as ai_proposed and is NEVER legally final until an authorized Revenue Officer signs off. Second, our Immutable Audit Log uses SHA-256 cryptographic hash chaining (blockchain principle). It records the exact timestamp, surveyor officer ID, legal justification notes, and before/after coordinates. It satisfies Section 65B of the Indian Evidence Act for electronic records.'

Q: Judge: 'What if there is no Survey of India CORS station in a rural village?'

A: Answer: 'Formula 8.4 handles missing ground truth gracefully. If no CORS station is within range, the GNSS term assigns a neutral penalty of 50.0 (moderate uncertainty, not zero). The parcel's priority score will then be driven primarily by the legacy record conflict (35%) and model segmentation confidence (30%). It never breaks.'

Q: Judge: 'How does this integrate into the government's existing DILRMP / BhuNaksha portal?'

A: Answer: 'NAKSHA-AI does not try to replace BhuNaksha. It acts as an intelligent pre-processing and verification triage layer. Our backend exposes standard OGC-compliant GeoJSON, Shapefiles, and REST APIs that plug directly into NIC's BhuNaksha portal and DILRMP state databases.'

9. Tech Stack & Engineering Architecture

NAKSHA-AI is architected with modern, enterprise-grade open-source software:

Layer	Technology Used	Key Responsibility
Backend API	FastAPI (Python 3.12/3.14)	Async REST endpoints, Pydantic v2 schemas, live WebSockets (/ws/live-queue).
Spatial Engine	Shapely & pyproj	Metric geodetic reprojection (UTM 43N), IoU overlap, discrete Hausdorff distance.
Web-GIS UI	React 19 + TypeScript + Leaflet	Interactive multi-layer GIS canvas, polygon vertex dragging, dark aesthetic theme.
Styling & Icons	TailwindCSS + Lucide React	Glassmorphism panels, responsive layouts, color-coded status badges.
Containers	Docker & Docker Desktop	Unified production container serving backend, API, and frontend on port 8000.
Storage Policy	D: Drive Isolation	D:\naksha_env, D:\naksha_cache, D:\naksha_data preserving C: drive disk space.

10. Glossary of Indian Land Revenue & GIS Terms

Use these authentic revenue terms when presenting to impress government evaluators and DoLR judges:

Term	Traditional / Technical Meaning	Context in NAKSHA-AI
Naksha / Shajra	Traditional village cadastral map showing plot boundaries.	The core map digitized and updated by NAKSHA-AI from drone imagery.
Khasra Number	Unique parcel identity number assigned to each plot in revenue records.	Every parcel in NAKSHA-AI carries its legal Khasra ID (e.g., Khasra 129/4B).
Jamabandi (RoR)	The official Record of Rights document containing ownership and plot area.	Our 'Legacy Cadastral Layer' against which AI boundaries are compared.
SVAMITVA	Flagship Ministry of Panchayati Raj drone survey scheme for rural Abadi land.	NAKSHA-AI uses Survey of India's SVAMITVA geodetic CORS network as ground truth.
CORS Network	Continuously Operating Reference Stations installed by Survey of India.	Ground truth satellite anchor pillars with ±1cm accuracy that never lie.
IoU (Overlap Index)	Intersection-over-Union: metric measuring shape agreement (0.0 to 1.0).	Measures geometric alignment between AI and legacy records (1.0 = perfect match).
Hausdorff Distance	The greatest physical gap (in real meters) between two boundary shapes.	Surfaces real-world encroachments (e.g. boundary wall drifted 6.8m outward).
Planar Graph	Cadastral topological rule: polygons must fit like puzzle pieces.	Enforced in our pipeline so parcels never overlap and leave no sliver gaps.

DOCUMENT VALIDATION & SYSTEM VERIFICATION:

- **Automated Test Suite:** 10 out of 10 backend tests passing (Pytest verified).
- **Production Bundle:** 1,907 TypeScript modules compiled in 6.7s with 0 errors.
- **Docker Desktop:** Unified container image naksha-ai:latest verified operational.
- **Dataset Specification:** Grounded on Survey of India CORS SVAMITVA datum and GSDL/OSM urban footprints.
- **API Key Requirement:** ZERO API keys required. 100% free and open-source out of the box.